

# Gappy AI Hackathon: Dual-Persona Command Center Blueprint

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## EXECUTIVE SUMMARY

This report outlines the technical and functional blueprint for the 'Dual-Persona Work & Study Command Center' project developed for the Gappy AI Hackathon. It details the user journey through a dynamic React single-page application, explaining the real-time data ingestion process via the Lemma SDK. Core features for both Student (Academic) and Professional (Enterprise) workspaces are specified, emphasizing tailored tools like attendance trackers, Kanban boards, and project management timelines. The document highlights a robust system design focused on personalized user experience and real-time data synchronization.

## User Journey and Website Flow

The application leverages a dynamic React single-page application (SPA) architecture, orchestrated with a `wouter` router, to provide a seamless user experience. The journey begins with flexible authentication options, allowing users to log in via OAuth (Google/GitHub) or opt for a convenient guest mode. Post-authentication, users are directed to an onboarding selector where they define their primary persona: 'Student' or 'Professional.' This selection dynamically customizes their dashboard experience.

The core of the application's functionality revolves around a real-time ingestion loop, powered by the Lemma SDK. This process starts locally with a user script that records audio, compresses it, and saves formatted notes. The server-side Lemma File Watcher instantly detects these new files, routing their content to the Lemma Triage Agent. This agent intelligently parses the data and updates the Lemma Datastore, triggering live refreshes on the user's frontend dashboard to reflect changes in real-time on features like Kanban boards and timelines.

- Dynamic React SPA offers a responsive and intuitive user interface.
- Dual authentication methods (OAuth & Guest Mode) enhance accessibility.
- Real-time data ingestion via Lemma SDK ensures dashboards are always up-to-date.
- Personalized dashboards are dynamically rendered based on user persona selection.

# Core Feature Specifications: Dual-Persona Workspaces

The 'Dual-Persona Work & Study Command Center' is distinguished by its two tailored dashboard workspaces, each designed to meet the specific needs of its target user group. The **Student Workspace**, branded as 'Academic Mode,' offers a 'Fluid Class Timeline' for tracking academic events with interactive attendance toggles. A '75% Attendance Guard' visually alerts students if their attendance falls below critical thresholds, complemented by a dedicated 'Academic Kanban Board' filtered by tags like `ACADEMICS` and `HARDWARE\_DEV`. Furthermore, students benefit from a 'Hardware Project Tracker' for managing circuit builds and a 'CGPA Simulator' to project academic performance.

The **Working Professional Workspace**, or 'Enterprise Mode,' provides an equally specialized suite of tools. It features a 'Sprint Timeline' designed to map daily standups, client demos, sprint reviews, and deployment windows, aligning with agile development methodologies. This workspace also incorporates a 'Billable Hours & Quorum Ledger,' crucial for professionals to track time and engagement efficiently. While the description of this workspace was cut short, its initial features clearly aim at enterprise-level project and time management.

- Two distinct dashboards cater to 'Student' and 'Professional' personas, offering highly specialized tools.
- Student Workspace features include academic timelines, attendance tracking, a specialized Kanban board, hardware project management, and a CGPA simulator.
- Professional Workspace focuses on agile methodologies with sprint timelines and essential billable hour tracking.
- The feature set for each persona is designed to integrate seamlessly with the real-time data ingestion system.

## Conclusion & Recommendations

The Dual-Persona Work & Study Command Center represents an innovative solution for dynamic, personalized productivity and management. By meticulously crafting two distinct yet integrated workspaces, the project effectively addresses the unique demands of both academic and professional users. The robust technical architecture, featuring a React SPA, real-time audio file ingestion via the Lemma SDK, and a responsive datastore, ensures that users receive timely and accurate updates, fostering an environment of efficiency and informed decision-making.

Key strengths lie in its adaptable design, real-time data processing capabilities, and the granular control offered through features like attendance tracking, academic Kanban boards, and sprint management. Future enhancements could explore integrating AI-powered insights from the 'Lemma Triage Agent' to offer proactive recommendations, expanding the hardware project tracker with inventory management, or adding collaborative features within the professional workspace. Overall, this hackathon project establishes a strong foundation for a comprehensive command center poised to empower users across diverse work and study contexts.